

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

Answer

- A. **130**
- B. **63**
- C. **4**
- D. **0**

Correct Answer: C

Rationale

Choice C is correct. It's given that the geologist has already collected **63** samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least **67** samples. Therefore, $63 + x \geq 67$. Subtracting **63** from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of **4** additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects **0** additional samples, the geologist will have collected a total of **63** samples, which is less than **67** samples.